Retained native session exporter: implementation and cache diagnosis

Task T-1499. 3 November 2026.

Implementation	Native postrun custody and typed joins
Validation failure	Cached graph compiler identity mismatch
Historical probes	Three rejected; no tests executed
Source rebuild	Graph built; native library LLVM allocation failed

Ordinary software evidence only. Prior failed attempts retained. T-1499.

The source implements a postrun exporter through the existing FreshOwnerPins custody check, SessionResourceJournal::close_fresh_enclosing_evidence, and the native diagnostic_file_adapter::check_file typed-session validator. It snapshots bounded native records and retains independently supplied controller receipt digest and owner pins. Self-asserted JSON flags do not authenticate controller custody. Worker allocation, active-worker time and exact kernel exit remain null; scientific admission and mathematical row acceptance remain false.

The intended source controls cover complete synthetic record joins, mismatched expected identities, mutated native files, owner lifecycle errors, changed PID start identity against the supplied digest, self-asserted authentication, prelaunch records, and symlink rejection. These controls have not run. Compilation failed before tests, so the source remains unverified and requires successful software validation and a non-author source review.

The first authorized 120-second, hard 2 GiB-address-space attempt failed after 4.153430 seconds with E0463 loading cached crypto_lib. All 341 pinned inputs were unchanged and children were reaped. A subsequent separately frozen 60-second metadata-only diagnostic failed after 5.165568 seconds with the precise E0460 dependency mismatch: crypto_lib expects pencil_symbolic_graph SVH 11e2bf11f1c6bb14375851999954ac1d; the current graph exposes b6b739a4e0323ebbcf9464ceb3626837. This identifies incompatible compiled metadata; it does not support increasing memory to repair that error.

A second separately frozen 60-second diagnostic checked three historical graph pairs in isolated symlink search directories, stopping only if a complete metadata load succeeded. All three were rejected in 3.720699 seconds: n83-r38 exposed eb9b3b72d8bed76cf254de622a9de45a, ic-wide exposed 461beef961343ec7eeccc1b1f745e49e, and ml-prime-root exposed 2bdbc89e7a5adcb8e3325915da632461. All 470 inputs were unchanged, all three children reaped, and no tests ran. The three ordinary validation envelopes total 240 seconds; their recorded elapsed time totals 13.039697 seconds. No scientific workload or cloud operation was invoked.

The static Cargo fingerprint audit found 289 nodes and no missing dependencies, but that was insufficient to establish embedded compiler metadata compatibility. Alternate cache paths and timestamps were selection evidence only. Cached compiled dependencies are not described as freshly rebuilt or source-attested. Prior failures and exact logs are retained rather than replaced by the later diagnosis.

The separately authorized coherent source build materialized exact source from commit c8db18e2c57e898d8b9a9b5097ada06254b32f08: 1103 source files totaling 59735879 bytes, plus 505 pinned read-only compiled/native inputs totaling 795512641 bytes. The graph library compiled successfully. The crypto_lib compiler then aborted with exit -6 and the exact message “rustc-LLVM ERROR: out of memory / Allocation failed” under the hard 8 GiB address-space limit. The attempt lasted 73.820523 seconds within its separate 300-second envelope. Both compiler children were reaped, no signal remained pending, and no exporter tests ran. No crypto_lib rlib was produced; partial metadata is retained as failure evidence.

The output monitor observed at most 169391025 bytes and at least 20030656512 bytes of free filesystem space. These are sampled observations; the accepted 4 GiB output threshold was a monitor rather than a filesystem quota. A separate read-only postfailure check verifies the 1608 source/reused inputs because the runner’s postcompile check was reached only on success. All four ordinary validation envelopes total 540 seconds; recorded elapsed time totals 86.860220 seconds. No retry follows this exhausted source-build allowance. The exporter source remains implemented but untested; completing a compatible native library build is its immediate validation dependency.

Primary references reviewed: Rust compiler command-line arguments, compiler debugging/tracing documentation, and Linux getrlimit semantics. Existing native source and the retained cohort-gates NEXT.md supply the API and postrun-order contract. The report is authored by the exporter producer; independent review remains required.